

Nucleophilic substitution proceeds through a trigonal bipyramidal transition state. The nucleophile attacks the carbon from the face opposite the leaving group, and inversion of configuration follows. This is the S_N2 mechanism, and its rate depends on the concentration of both the substrate and the nucleophile. Steric bulk at the reacting carbon slows it dramatically, which is why tertiary halides do not undergo S_N2 at any useful rate.